Brute-force Complexity

If $x^2+y^2=n$,
then
$|x| \leq \sqrt{n}$ and $|y| \leq \sqrt{n}$

Therefore, the brute-force method only needs to check integer pairs s.t.

$$-\lfloor\sqrt{n}\rfloor \leq x \leq \lfloor\sqrt{n}\rfloor$$
$$-\lfloor\sqrt{n}\rfloor \leq y \leq \lfloor\sqrt{n}\rfloor$$

There are approx $2\sqrt{n}+1$ possible values of x and $2\sqrt{n}+1$ values of y.

So the total no. of pairs checked is:

$$(2\sqrt{n}+1)^2$$

For large n, the '+1' is negligible
Thus,

$$(2\sqrt{n})^2 = 4n.$$